2026 CYBER SITH WHITE TEAM DOCUMENTATION

(Team 13, Urbandale Highschool)

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Network Architecture

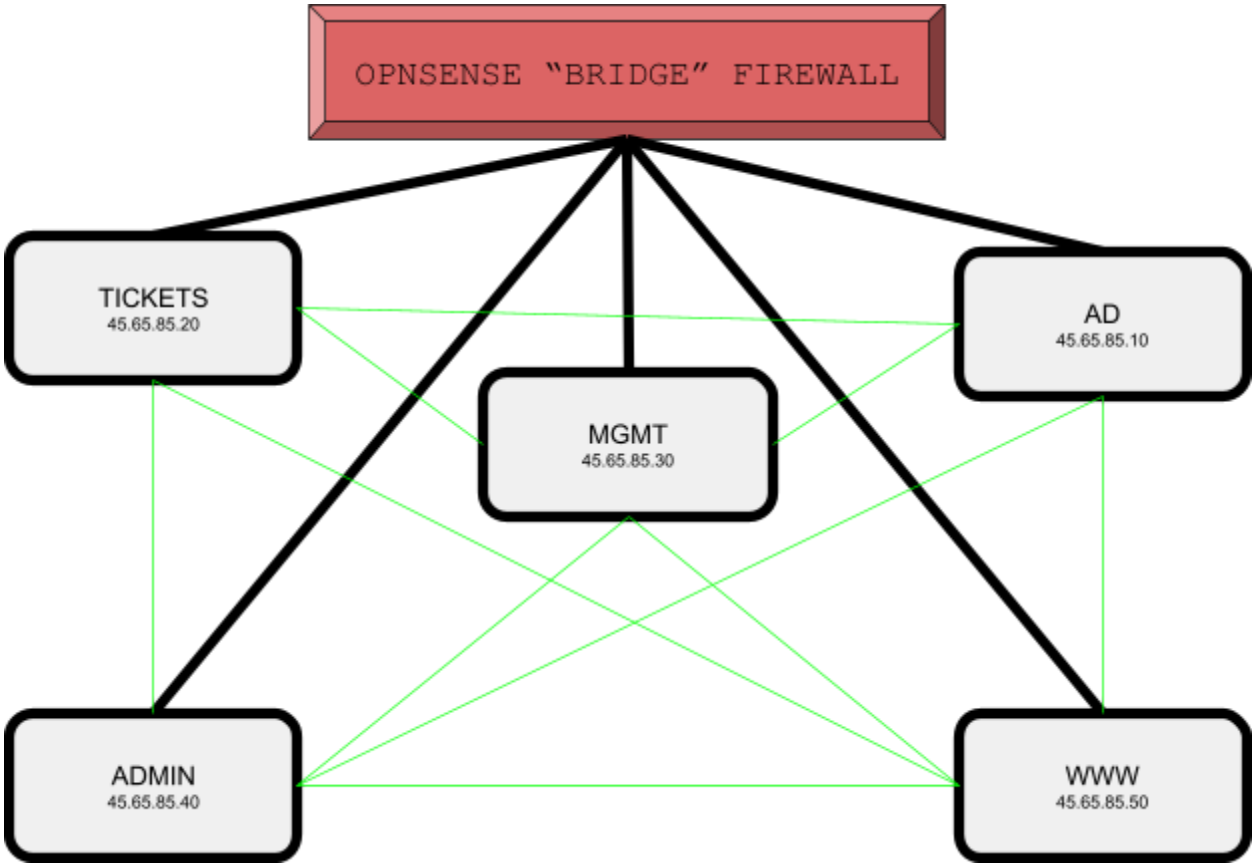
For our network architecture we created a bridged firewall with OPNSense to protect our servers without changing or modifying the IP addresses of the servers provided. This meant all "horizontal" traffic from outside our network would be forced through our strict firewall and traffic inside our system would be contained. This way our servers could communicate and function un-impeeded between one and other, while still being protected from outside threats.

Publicly Accessible Services

Service Domain Name	Service IP Address
ad.team13.isucdc.com	45.65.85.10
tickets.team13.isucdc.com	45.65.85.20
mgmt.team13.isucdc.com	45.65.85.30
admin.team13.isucdc.com	45.65.85.40
www.team13.isucdc.com	45.65.85.50

These are all the protected and provided services within our network.

Network Topology Map



The image above shows how our network is structured, with all inputs coming through the firewall to their destinations and all servers behind the firewall being able to communicate unimpeded. Requiring all inward and outward traffic to pass through our firewall and thus be subjected to its WAN rules ensures the information and access is more secure. Given the scenario presented, we felt this was the best course of action for defending our services. Our OPNSense firewall also requires information pass back through it

OPNSense Firewall

Our firewall employs OPNSense as a blanket firewall over all of our machines because of its accessibility and usability for us and our services. It appealed to us for a number of reasons.

- The usability of the service made it easy to implement rules and create a bridged Firewall for our services.
- It avoided the need to change the IP address of our services and thus lessened the workload for the hardening and securing of our system
- It offers live log file viewing for easy monitoring of incoming and outgoing traffic

Rules

WAN Rules

Only these ports were allowed through our firewall using WAN rules, while all others would be blocked from accessing our network. This allowed us to drastically limit the routes of attack and secure all the servers without compromising internal communication. These were all allowed on IPv4 TCP to allow a stronger control over access.

	IPv4 TCP	*	*	45.65.85.10	636	*	*	AD LDAPS	
	IPv4 TCP	*	*	45.65.85.10	3389 (MS RDP)	*	*	AD RDP	
	IPv4 UDP	*	*	45.65.85.10	53 (DNS)	*	*	AD DNS	
	IPv4 TCP	*	*	45.65.85.20	5000	*	*	Ticketing HTTP	
	IPv4 TCP	*	*	45.65.85.20	3389 (MS RDP)	*	*	Ticketing RDP	
	IPv4 TCP	*	*	45.65.85.40	80 (HTTP)	*	*	Admin HTTP access	
	IPv4 TCP	*	*	45.65.85.40	3389 (MS RDP)	*	*	Admin RDP access	
	IPv4 TCP	*	*	45.65.85.30	5000	*	*	MGMT HTTP	
	IPv4 TCP	*	*	45.65.85.30	22 (SSH)	*	*	Management SSH	
	IPv4 TCP	*	*	45.65.85.50	80 (HTTP)	*	*	www http	
	IPv4 TCP	*	*	45.65.85.50	22 (SSH)	*	*	www ssh	
	IPv4 TCP	199.100.16.100	*	*	*	*	*		
	IPv4 ICMP	199.100.16.100	*	45.65.85.0/24	*	*	*		

OPT1 Rules

These "Optional rules” create exceptions to our firewalls mass restriction on port traffic and allow our network to function through limited communications to the outside on these ports.

☐	▶	→	ϟ	ℹ	IPv4 TCP	*	*	199.100.16.100	3128	*	*
☐	▶	→	ϟ	ℹ	IPv4 UDP	*	*	199.100.16.100	53 (DNS)	*	*
☐	▶	←	ϟ	ℹ	IPv4 UDP	45.65.85.10	*	199.100.16.100	53 (DNS)	*	*
☐	▶	→	ϟ	ℹ	IPv4 *	OPT1 net	*	OPT1 net	*	*	*
☐	▶	→	ϟ	ℹ	IPv4 UDP	OPT1 net	*	*	53 (DNS)	*	*

Servers

Server Hardening

In order to protect our servers from unwanted intrusion we also took steps to ensure we remained protected on the software end of our network. The steps taken were:

- **Root and Server Login:** All of the default login and root passwords were changed on each machine provided to much more secure and notably unknown passwords to ensure the most obvious point of entry is inaccessible.
- **Accounts and Permissions:** A number of the linux machines have unnecessary and unwanted accounts with permissions that would compromise servers and data. These accounts were all disabled through either rules that wouldn't permit their access to the server or measures like removing permissions and changing login passwords.
- **Preset vulnerabilities and exploits:** There were a number of vulnerabilities created and hidden on the machines provided that were removed due to their compromising nature:
    - Enabled and restored windows Defender
    - Removed a secluded program that would have reset passwords
    - Removed a GitHub program in our windows machines browser
- **Windows Content Trimming:** All of the provided windows servers came with applications and processes that were unnecessary and exploitable for our system. The following services were uninstalled and removed from our windows

machines alongside their ports being disabled: Clip Champ, Microsoft News, Xbox, Xbox Live, Solitaire, Sticky Notes, Microsoft Movies, Microsoft People.

Servers

AD

This server manages authentication and thus is very important to the security of our network as a whole. Due to its importance, this server was refined and examined to ensure our authentication process worked as intended and was as strict and secure as possible.

Tickets

This is the most important part of our system for data storage and is integral to the service we provide for users. Fortunately there is little to exploit or manipulate on this server in terms of injection or users so it should remain secure.

MGMT

This server works as the backend of a lot of our services and thus received extra attention when hardening. The files and code of this linux machine were examined thoroughly and we were able to remove unwanted users as well as eliminate a hazardous program associated with root permutations.

ADMIN

This server is used for the cities admin portal that displays the statuses of the cities infrastructure. For this we ensure our user permissions and credentials were changed and updated to ensure that no unwanted entry was allowed.

WWW

This is the most integral piece of our user interface and thus is the most exploitable. We took care to strictly regulate ports and permissions within this machine in order to avoid any obvious and easy exploits such as XSS or SQL injection. We also worked to ensure that users on this machine would only have the absolutely necessary permissions to ensure that as little damage as possible could be done in the case of a breach or compromise to our security.